$$\hat{q}_i = \sqrt{\frac{1}{2m_i\omega_i}}\left(\hat{a}_i + \hat{a}_i^\dagger\right), \tag{5.89a}$$

$$\hat{p}_i = -\mathrm{i}\sqrt{\frac{m_i\omega_i}{2}}\left(\hat{a}_i - \hat{a}_i^\dagger\right), \tag{5.89b}$$